

Integrated In Silico Characterization of the Myostatin (MSTN/GDF8) Gene in Livestock Animals: Sequence Analysis, Evolution, Structure Prediction, and Molecular Docking

To retrieve and curate nucleotide and protein sequences of the MSTN (GDF8) gene from publicly available biological databases for cattle and selected livestock species.

Abstract

The MSTN gene codes for myostatin protein. This protein is part of the transforming growth factor beta (TGF β) superfamily, which is a group of proteins that help control the growth and development of tissues throughout the body. It is found exclusively in muscles. It is active before and after birth and usually limits muscle growth to regulate muscle growth, differentiation and tissue repair. It is actively studied in mice, cows, and other animals. It is said to have similar function in humans.

Some cattle breeds like Belgian Blue Cattle have naturally occurring MSTN genes. Thus, it allows us to study how to increase muscle mass and lean meat production.

Researchers are studying myostatin as a potential treatment for various muscular dystrophies that cause muscle weakness and wasting (atrophy).

This project focuses in analysing the MSTN gene and protein in *Bos taurus* using in silico approaches to understand its sequence, structure, evolution, and interactions with bioactive compounds, thereby assessing its role in muscle development and its potential applications in animal breeding and biotechnology.

Myostatin, MSTN, GDF8, Bioinformatics, Molecular Docking, AlphaFold, Livestock Genetics, Protein Structure, Comparative Genomics, SwissDock

The following sequences were downloaded from publicly downloaded databases: -

gene.fna : consists the complete genomic sequences

rna.fna : consists of mature rna transcript

cds.fna : consists of the coding DNA sequence

protein.faa : consists of the protein sequence

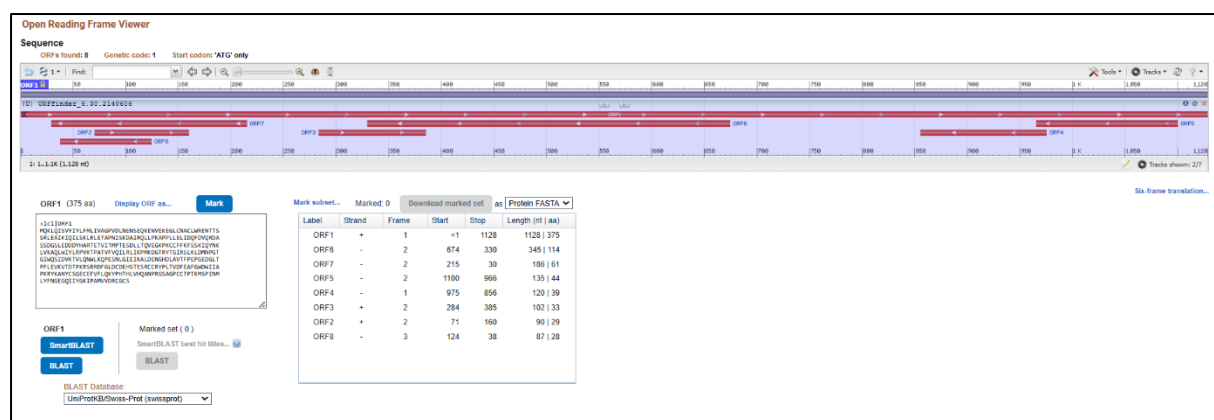
gene_data_table.tsv : contains geneID and description of datasets

gene_product_table.tsv : contains transcript IDs, proteins IDs used to report accession numbers.

To identify the coding sequence and open reading frame (ORF) of the MSTN gene and analyze its basic sequence characteristics.

We first retrieve the CDS since it is the exact DNA sequence that gets translated into myostatin protein. We also find the ORF which tells us where the protein starts and ends confirming that the coding sequence can produce complete functional protein.

We paste the cds sequence in the NCBI ORFfinder; the results are as follows: -



Mark subset... Marked: 0 Download marked set as Protein FASTA

Label	Strand	Frame	Start	Stop	Length (nt aa)
ORF1	+	1	<1	1128	1128 375
ORF6	-	2	674	330	345 114
ORF7	-	2	215	30	186 61
ORF5	-	2	1100	966	135 44
ORF4	-	1	975	856	120 39
ORF3	+	2	284	385	102 33
ORF2	+	2	71	160	90 29
ORF8	-	3	124	38	87 28

From the result, we notice that: -

Total ORFs found: 8

Correct ORF: ORF1

Strand: + (Forward)

Start position: <1

Stop position: 1128

ORF length: 1128 bp

Protein length: 375 amino acids

We choose ORF1 since it is the longest ORF (1128bp), located in the forward strand (+), starts at the beginning of CDS, encodes a 375 amino acid protein that matches the known length of the MSTN precursor protein.

Thus, it is an uninterrupted coding sequence. It produces 375 precursor protein and a stop codon. The remaining ORFs are considerably shorter and unlikely to represent functional proteins. Identification of the correct ORF confirms the integrity of the MSTN coding sequence and provides the appropriate amino acid sequence for analysis.

Next the A, T, G, C content using the sequence manipulation suite. The results are as follows:

DNA Stats results		
Results for 1128 residue sequence "NM_001001525.4:134-1261 MSTN [organism=Bos taurus] [GeneID=281187] [region=cds]" starting "ATGCAAAAAC"		
Pattern:	Times found:	Percentage:
g	249	22.07
a	352	31.21
t	287	25.44
c	240	21.28
n	0	0.00
u	0	0.00
r	0	0.00
y	0	0.00
s	0	0.00
w	0	0.00
k	0	0.00
m	0	0.00
b	0	0.00
d	0	0.00
h	0	0.00
v	0	0.00
gg	55	4.88
ga	94	8.34
gt	48	4.26
gc	52	4.61
gn	0	0.00
ag	77	6.83
aa	120	10.65
at	87	7.72
ac	67	5.94
an	0	0.00
tg	98	8.70
ta	57	5.06
tt	74	6.57
tc	58	5.15
tn	0	0.00

cg	19	1.69
ca	80	7.10
ct	78	6.92
cc	63	5.59
cn	0	0.00
ng	0	0.00
na	0	0.00
nt	0	0.00
nc	0	0.00
nn	0	0.00
g,c	489	43.35
a,t	639	56.65
r,y,s,w,k	0	0.00
b,h,d,v,n	0	0.00
r,y,s,w,k,m,b,d,h,v,n	0	0.00

Parameter	Value
Sequence length	1128 bp
A	352 (31%)
T	287 (25.44%)

G	249 (22.07%)
C	240 (21.28%)
GC content	43.35 %
AT content	56.65 %

This indicates that MSTN sequence is AT rich, and it has a moderate GC content. A moderate GC content contributes to a good DNA stability. GC-rich regions have higher melting temperatures because G-C pairs form three hydrogen bonds, while A-T pairs form only two. It is also suitable for PCR and allows us to replicate the DNA sequence allowing more production of myostatin protein. A higher AT – rich sequence influence DNA flexibility, transcription efficiency and a stable coding sequence.

To evaluate sequence similarity and evolutionary conservation of MSTN among different livestock species using BLAST analysis.

Now we operate the BLAST (Basic local alignment search tool) against sequences in NCBI database to compare if it's this really the MSTN gene, which species have similar MSTN genes, how similar are they, has the gene been conserved during evolution.

To compare the genes, we use the file cds.fna The NCBI blast website is used to perform BLAST; specifically, BLASTN since we are comparing DNA sequences.

The screenshot displays the NCBI BLASTN web interface. The top section, 'Enter Query Sequence', includes a text area with the sequence: CTATATTTTAAATGGCGAAGGACAAATAATATACGGGAAGATTCCAGCCATGG TAGTAGATCGCTGTGGGT GTTCATGA. Below this is a 'Job Title' field containing 'NM_001001525.4:134-1261 MSTN [organism=Bos...]' and a checkbox for 'Align two or more sequences'. The 'Choose Search Set' section allows selecting 'Standard databases (nr etc.)' and 'Core nucleotide database (core_nt)'. It also includes options to 'Exclude' models, uncultured sequences, and type material, and a 'Limit to' field for Entrez queries. The 'Program Selection' section at the bottom shows 'Optimize for' with radio buttons for 'Highly similar sequences (megablast)', 'More dissimilar sequences (discontiguous megablast)', and 'Somewhat similar sequences (blastn)'. The interface is titled 'BLASTN programs search nucleotide databases using a nucleotide query. more...'

The results are as follows:-

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
✓	Bos taurus myostatin (MSTN). mRNA	Bos taurus	2084	2084	100%	0.0	100.00%	2768	NM_001001525.4
✓	Bos taurus myostatin (GDF8) mRNA, complete cds	Bos taurus	2084	2084	100%	0.0	100.00%	1128	AY160688.1
✓	Bos taurus myostatin mRNA (cDNA clone MGC:143019 IMAGE:8308041). complete cds	Bos taurus	2084	2084	100%	0.0	100.00%	2738	BC134563.1
✓	PREDICTED: Bos mutus myostatin (MSTN). mRNA	Bos mutus	2078	2078	100%	0.0	99.91%	2730	XM_005907427.2
✓	Bos taurus myostatin mRNA, complete cds	Bos taurus	2078	2078	100%	0.0	99.91%	1140	GQ184147.1
✓	Bos taurus myostatin (MSTN) mRNA, complete cds	Bos taurus	2078	2078	100%	0.0	99.91%	1128	AF019620.1
✓	PREDICTED: Bison bison bison myostatin (MSTN). mRNA	Bison bison bison	2078	2078	100%	0.0	99.91%	2762	XM_010862903.1
✓	PREDICTED: Bos javanicus myostatin (MSTN). mRNA	Bos javanicus	2073	2073	100%	0.0	99.82%	2832	XM_061432883.1
✓	PREDICTED: Bos indicus x Bos taurus myostatin (MSTN). mRNA	Bos indicus x Bos taurus	2067	2067	100%	0.0	99.73%	2838	XM_027555492.1
✓	PREDICTED: Bos indicus myostatin (MSTN). mRNA	Bos indicus	2061	2061	100%	0.0	99.65%	2887	XM_019969486.2
✓	Bos taurus myostatin (GDF8) mRNA, complete cds	Bos taurus	2045	2045	100%	0.0	99.38%	1128	AF019761.1
✓	Bubalus bubalis myostatin (MSTN). mRNA	Bubalus bubalis	2017	2017	100%	0.0	98.94%	1231	NM_001290967.1
✓	PREDICTED: Bubalus kerabau myostatin (MSTN). mRNA	Bubalus kerabau	2001	2001	100%	0.0	98.67%	2703	XM_055574715.1
✓	Bubalus bubalis myostatin (Mstn) mRNA, complete cds	Bubalus bubalis	1989	1989	100%	0.0	98.49%	1128	KP057613.1
✓	Taurotragus derbianus myostatin mRNA, complete cds	Taurotragus derbianus	1967	1967	100%	0.0	98.14%	1128	AY629304.1
✓	PREDICTED: Moschus berezovskii myostatin (MSTN). mRNA	Moschus berezovskii	1934	1934	100%	0.0	97.61%	2536	XM_055397770.1
✓	PREDICTED: Giraffa tippelskirchi myostatin (MSTN). mRNA	Giraffa tippelskirchi	1906	1906	100%	0.0	97.16%	2863	XM_081579306.1
✓	PREDICTED: Muntiacus reevesi myostatin (MSTN). mRNA	Muntiacus reevesi	1895	1895	100%	0.0	96.99%	2573	XM_065927948.1
✓	PREDICTED: Dama dama myostatin (MSTN). mRNA	Dama dama	1890	1890	100%	0.0	96.90%	1994	XM_061135030.1
✓	PREDICTED: Cervus elaphus myostatin (MSTN). mRNA	Cervus elaphus	1884	1884	100%	0.0	96.81%	2761	XM_043894881.1
✓	PREDICTED: Cervus canadensis myostatin (MSTN). mRNA	Cervus canadensis	1879	1879	100%	0.0	96.72%	2761	XM_043488279.1

✓	Aepyceros melampus myostatin mRNA, complete cds	Aepyceros melampus	1873	1873	100%	0.0	96.63%	1128	AY629307.1
✓	PREDICTED: Odocoileus virginianus myostatin (MSTN). mRNA	Odocoileus virginianus	1868	1868	100%	0.0	96.54%	2657	XM_020900528.2
✓	PREDICTED: Budorcas taxicolor myostatin (MSTN). mRNA	Budorcas taxicolor	1862	1862	100%	0.0	96.45%	1279	XM_052635569.1
✓	Cervus elaphus xanthopygus myostatin mRNA, complete cds	Cervus elaphus xanthopygus	1862	1862	100%	0.0	96.45%	1331	EF629535.1
✓	PREDICTED: Oryx dammah myostatin (MSTN). mRNA	Oryx dammah	1857	1857	100%	0.0	96.37%	4162	XM_040227852.1
✓	PREDICTED: Ovis canadensis myostatin (MSTN). mRNA	Ovis canadensis	1857	1857	100%	0.0	96.37%	2770	XM_069577003.2
✓	Ovis aries myostatin (MSTN) mRNA, complete cds	Ovis aries	1857	1857	100%	0.0	96.37%	1128	AF019622.1
✓	Capra ibex myostatin mRNA, complete cds	Capra ibex	1857	1857	100%	0.0	96.37%	1128	AY629305.1
✓	Ovis aries myostatin (MSTN). mRNA	Ovis aries	1857	1857	100%	0.0	96.37%	2894	NM_001009428.3
✓	Ovis aries mRNA for myostatin (mstn gene), wild type allele	Ovis aries	1857	1857	100%	0.0	96.37%	1394	AM992883.1
✓	Capra hircus breed Qianbei Ma Goat myostatin mRNA, complete cds	Capra hircus	1857	1857	100%	0.0	96.37%	1128	JN662463.1
✓	Ovis aries growth differentiation factor 8 precursor (GDF8) mRNA, complete cds	Ovis aries	1851	1851	100%	0.0	96.28%	1647	KM371731.1
✓	Capra hircus breed Sirohi myostatin precursor (MSTN) mRNA, complete cds	Capra hircus	1851	1851	100%	0.0	96.28%	1250	HM462259.1
✓	Capra hircus myostatin (GDF8) mRNA, complete cds	Capra hircus	1851	1851	100%	0.0	96.28%	1128	AY436347.1
✓	Hemitragus jemlahicus myostatin mRNA, complete cds	Hemitragus jemlahicus	1851	1851	100%	0.0	96.28%	1128	AY629306.1
✓	PREDICTED: Capricornis sumatraensis myostatin (MSTN). mRNA	Capricornis sumatraensis	1851	1851	100%	0.0	96.28%	1128	XM_068968342.1
✓	Capra hircus myostatin precursor mRNA, complete cds	Capra hircus	1851	1851	100%	0.0	96.28%	1128	JN639006.1
✓	Capra hircus myostatin (MSTN). mRNA	Capra hircus	1851	1851	100%	0.0	96.28%	1255	NM_001285737.1
✓	Capra hircus breed Black Bengal myostatin precursor (MSTN) mRNA, complete cds	Capra hircus	1851	1851	100%	0.0	96.28%	1250	HM462261.1
✓	Ovis aries mRNA for myostatin (mstn gene), allele nt120-ins1	Ovis aries	1851	1851	100%	0.0	96.28%	1362	FM207636.1
✓	Capra hircus breed Guizhou White myostatin (MSTN) mRNA, complete cds	Capra hircus	1851	1851	100%	0.0	96.28%	1128	JX840482.1
✓	Antilocapra americana myostatin mRNA, complete cds	Antilocapra americana	1851	1851	100%	0.0	96.28%	1128	AY629309.1
✓	Ovis aries mRNA for myostatin (mstn gene), allele nt960-del1	Ovis aries	1849	1849	100%	0.0	96.28%	1393	AM992884.1

Query cover: Percent of MSTN sequence that aligns with database sequence.

Percent identity: Percentage of identical nucleotides between the sequences.

E-value: Probability that the match occurred by chance.

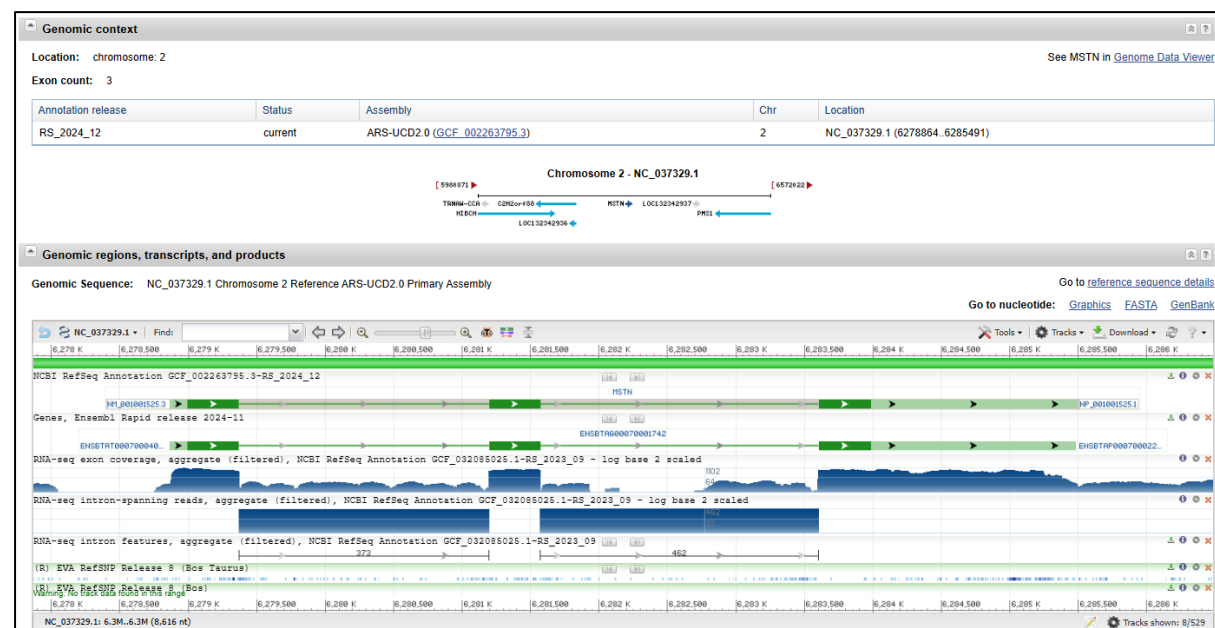
Max Score: Alignment quality score.

Thus, the BLASTN analysis of the *Bos Taurus* MSTN coding sequence identified highly significant homologous sequences amongst livestock species. All top 100 matches showed a 100% coverage indicating the entire coding sequence aligned with homologous MSTN genes sequence identity ranged from 98.94% to 100% among closely related livestock species, with *Bos mutus*, *Bison bison bison*, *Bos javanicus*, and *Bos indicus* exhibiting particularly high similarity to cattle. All alignments produced an E-value of 0.0, confirming that the observed similarities are statistically significant and not due to random chance. Thus, the MSTN gene is highly conserved in livestock species.

To predict and compare the exon–intron organization and genomic architecture of the MSTN gene.

This process helps us understand the genomic architecture of the MSTN gene. We query the NCBI gene and observe the summary.

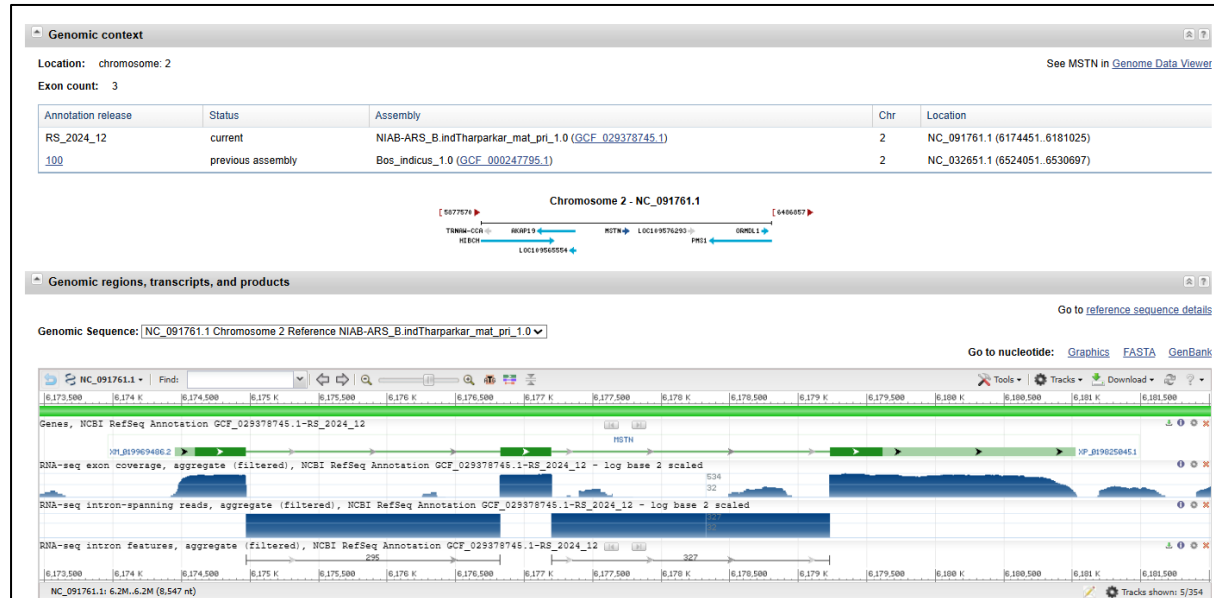
Parameter	Result
Gene Symbol	MSTN
Gene Name	Myostatin (GDF8)
Gene ID	281187
Organism	<i>Bos taurus</i>
Chromosome	2
Genomic Location	NC_037329.1 (6278864–6285491)
Exon Count	3



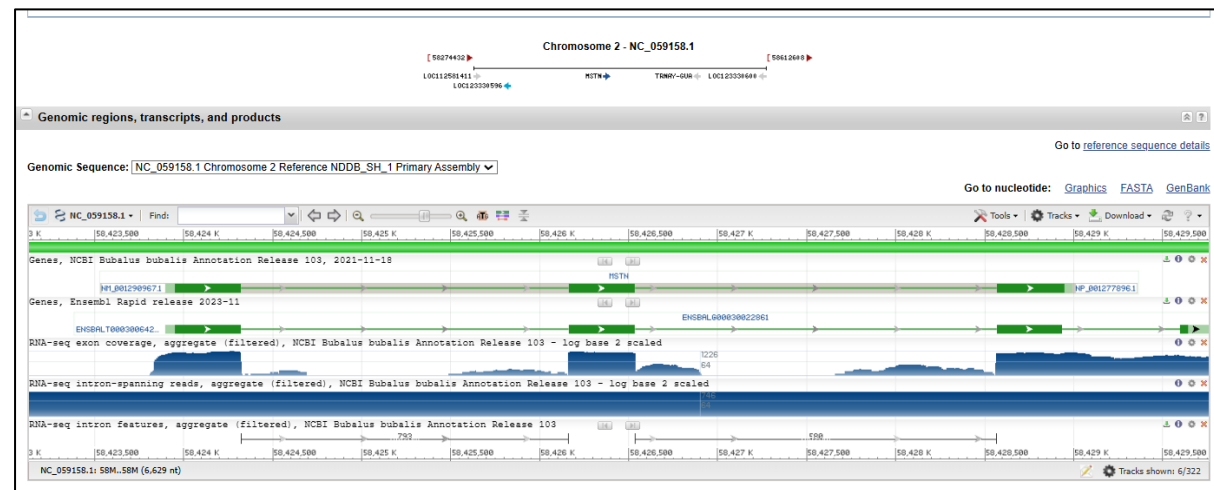
The MSTN gene is a protein coding gene that belongs to the transforming growth factor-beta (TGF- β) superfamily. It negatively regulates muscle growth. The MSTN gene is located on chromosome 2 of the *Bos taurus* genome and is approximately 5.6 kb (6,285,491 –

6,278,864 + 1 = 5,628 bp). The green boxes represent exons and the connecting lines represent introns. The arrow signifies direction of transcription. The conserved exon arrangement ensures proper RNA splicing and accurate synthesis of the myostatin protein.

These are the findings on comparing with similar species *Bos indicus* and *Bos bubalis*.



Genomic content of *Bos indicus*



Genomic content of myostatin *Bubalus bubalis*

Feature	<i>Bos taurus</i>	<i>Bos indicus</i>	<i>Bubalus bubalis</i>
Gene Symbol	MSTN	MSTN	MSTN
Gene Type	Protein-coding	Protein-coding	Protein-coding
Chromosome	2	2	2
Exon Count	3	3	3
Intron Count	2	2	2

Gene Organization	3 exons separated by 2 introns	3 exons separated by 2 introns	3 exons separated by 2 introns
-------------------	--------------------------------------	--------------------------------------	--------------------------------------

There is a highly conserved architecture amongst *Bos taurus*, *Bos indicus*, and *Bubalus bubalis*. In all three species, the gene is located in chromosome 2 and consists of 3 exons separated by 2 introns. Thus, it indicates strong evolutionary conservation.

To perform multiple sequence alignment for identifying conserved amino acid residues and functionally important regions of the MSTN protein.

It is performed to compare if amino acid content is identical amongst species, if it has changed and to check which parts of proteins are highly conserved. This signifies a lot about the evolution of MSTN gene. For this part, the following organisms would be referred to:-

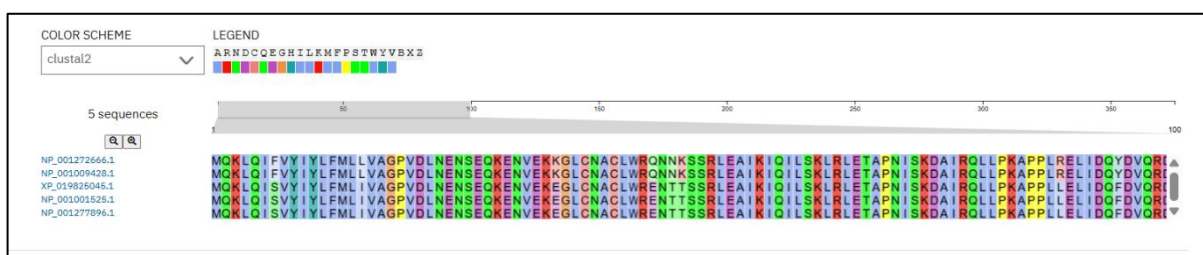
1. *Bos taurus*
2. *Bos indicus*
3. *Bubalus bubalis*
4. *Ovis aries*
5. *Capra hircus*

Conserved residues generally indicate functionally or structurally important regions of the protein, whereas variable residues reflect evolutionary divergence among species.

Protein sequences of MSTN from *Bos taurus*, *Bos indicus*, *Bubalus bubalis*, *Ovis aries*, and *Capra hircus* were retrieved from the NCBI Protein database. The sequences were aligned using Clustal Omega with the default parameters.

The multiple sequence alignment showed a high degree of amino acid conservation among all five livestock species. The document will be attached alongside.

Most positions throughout the 375-amino acid MSTN protein were marked by ‘*’, indicating identical amino acid residues across all species. Only a limited number of amino acid substitutions were observed, represented by ‘:’ and ‘.’ symbols. The region spanning approximately amino acids 181-240 exhibited complete conservation, with every residue marked by ‘*’, indicating an exceptionally conserved functional region. Similarly, the C-



terminal portion of the protein (approximately residues 241-375) also showed extensive conservation with only a few conservative substitutions.

NP_001272666.1	MQKLQIFVYIYLFMLLVAGPVDLNENSEQKENVEKGLCNACLRQNNKSSRLAIAIKI	60
NP_001009428.1	MQKLQIFVYIYLFMLLVAGPVDLNENSEQKENVEKGLCNACLRQNNKSSRLAIAIKI	60
XP_019825045.1	MQKLQISVYIYLFMLLVAGPVDLNENSEQKENVEKGLCNACLRNTTSSRLAIAIKI	60
NP_001001525.1	MQKLQISVYIYLFMLLVAGPVDLNENSEQKENVEKGLCNACLRNTTSSRLAIAIKI	60
NP_001277896.1	MQKLQISVYIYLFMLLVAGPVDLNENSEQKENVEKGLCNACLRNTTSSRLAIAIKI	60

NP_001272666.1	LSKLRLTAPNISKDAIRQLLPKAPPLRELIDQVDQRRDSSDGSLEDDDYHVTETVIT	120
NP_001009428.1	LSKLRLTAPNISKDAIRQLLPKAPPLRELIDQVDQRRDSSDGSLEDDDYHVTETVIT	120
XP_019825045.1	LSKLRLTAPNISKDAIRQLLPKAPPLLELIQFVQRRDSSDGSLEDDDYHARTETVIT	120
NP_001001525.1	LSKLRLTAPNISKDAIRQLLPKAPPLLELIQFVQRRDSSDGSLEDDDYHARTETVIT	120
NP_001277896.1	LSKLRLTAPNISKDAIRQLLPKAPPLLELIQFVQRRDSSDGSLEDDDYHARTDAVIT	120

NP_001272666.1	MPTESDLLAEVQEKPKCCFFKSSKIQHNVVKAQLWIYLRPVKTPPTVFVQILRLIKPM	180
NP_001009428.1	MPTESDLLAEVQEKPKCCFFKSSKIQHNVVKAQLWIYLRPVKTPPTVFVQILRLIKPM	180
XP_019825045.1	MPTESDLLTQVEGPKCCFFKSSKIQYNKLVKAQLWIYLRPVKTPATVFVQILRLIKPM	180
NP_001001525.1	MPTESDLLTQVEGPKCCFFKSSKIQYNKLVKAQLWIYLRPVKTPATVFVQILRLIKPM	180
NP_001277896.1	MPTESDLLTQVEGPKCCFFKSSKIQYNKLVKAQLWIYLRPVKTPATVFVQILRLIKPM	180

NP_001272666.1	KDGTRYTGIRSLKLDMPGTGIWQSDIVKTVLQNWMLKQPSNLGIEIKALDENGHD LAVT	240
NP_001009428.1	KDGTRYTGIRSLKLDMPGTGIWQSDIVKTVLQNWMLKQPSNLGIEIKALDENGHD LAVT	240
XP_019825045.1	KDGTRYTGIRSLKLDMPGTGIWQSDIVKTVLQNWMLKQPSNLGIEIKALDENGHD LAVT	240
NP_001001525.1	KDGTRYTGIRSLKLDMPGTGIWQSDIVKTVLQNWMLKQPSNLGIEIKALDENGHD LAVT	240
NP_001277896.1	KDGTRYTGIRSLKLDMPGTGIWQSDIVKTVLQNWMLKQPSNLGIEIKALDENGHD LAVT	240

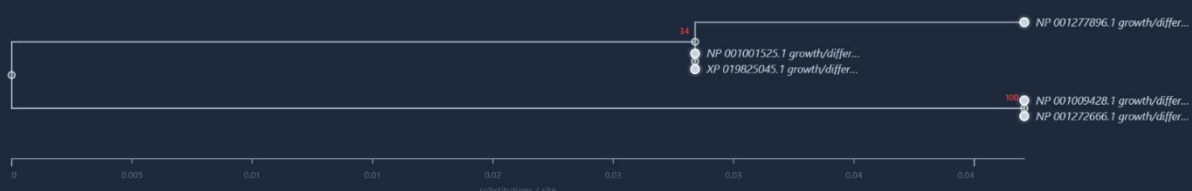
NP_001272666.1	FFPEGEEGLNPFLEVKVTDTPKRSRRDFGLDCDEHSTESRCCRYPLTVDFEAFGNDWIIA	300
NP_001009428.1	FFPEGEEGLNPFLEVKVTDTPKRSRRDFGLDCDEHSTESRCCRYPLTVDFEAFGNDWIIA	300
XP_019825045.1	FFPEGEDGLTPFLEVKVTDTPKRSRRDFGLDCDEHSTESRCCRYPLTVDFEAFGNDWIIA	300
NP_001001525.1	FFPEGEDGLTPFLEVKVTDTPKRSRRDFGLDCDEHSTESRCCRYPLTVDFEAFGNDWIIA	300
NP_001277896.1	FFPEGEDGLTPFLEVKVTDTPKRSRRDFGLDCDEHSTESRCCRYPLTVDFEAFGNDWIIA	300

Only minor variation occurred between the species. This extensive amino acid conservations shows that mutations are usually not tolerated as it disrupts its biological function. Since MSTN is a negative regulator of skeletal muscle growth. Thus, such regions are usually highly conserved for maintaining the correct protein structure, receptor binding, signal transduction, and regulation of muscle development. The completely conserved region observed between residues 181-240 suggests that this portion of the protein is

particularly important for MSTN function and has remained unchanged throughout livestock evolution

To infer the evolutionary relationships among livestock species based on MSTN gene/protein sequences through phylogenetic analysis

Phylogenetic analysis is performed to determine the evolutionary relationships among different livestock species based on the similarity of their MSTN sequences. Species with highly similar sequences are expected to cluster together, indicating a more recent common ancestor, whereas species with greater sequence differences are placed farther apart in the phylogenetic tree.



The fasta file of all MSTN sequences given above was uploaded to phylogeny.fr using the OneClick workflow. The platform automatically performed alignment refinement (Gblocks), phylogenetic tree construction using PhyML, and tree visualization using TreeDyn.

The phylogenetic tree groups the five organisms into two parts: -

1. *Bos taurus* and *Bos indicus* clustered together, indicating the highest sequence similarity and a very close evolutionary relationship.
2. *Bubalus bubalis* grouped with the cattle lineage but formed a separate branch, indicating that although buffalo is closely related to cattle, it has accumulated slight sequence divergence.
3. *Ovis aries* and *Capra hircus* formed a distinct cluster with a bootstrap value of 100, demonstrating very strong statistical support for their close evolutionary relationship.
4. The separation between the bovine group (*Bos* and *Bubalus*) and the caprine group (*Ovis* and *Capra*) reflects their divergence into different evolutionary lineages.

The clustering of *Bos taurus* with *Bos indicus* reflects the common ancestry within the genus *Bos* while the *Bubalus bubalis* on another branch is consistent with its classification of a different genus. *Ovis aries* and *Capra hircus* are closely related to the family *Caprinae*.

The bootstrap value of 100 for the *Ovis aries*-*Capra hircus* cluster indicates very high confidence in that grouping. The value 34 near the cattle/buffalo branch is relatively low, meaning the exact branching order among *Bos taurus*, *Bos indicus*, and *Bubalus bubalis* has weaker statistical support in this analysis.

To identify conserved sequence motifs and functional protein domains associated with MSTN using bioinformatics databases and motif discovery tools.

Motifs are short conserved amino acids patterns that help in function, stability and interactions. Protein domains are larger conserved structure responsible for biological activity of protein. Identifying these regions in the *Bos taurus* MSTN protein helps explain its role in regulating skeletal muscle development and provides insight into its structural organization.

The Fasta sequence of *Bos taurus* (protein sequence) was analysed using the InterProScan database. InterPro integrates multiple protein signature databases, including Pfam, SMART, PANTHER, PIRSF, PROSITE, and CDD, to identify conserved sequence motifs, protein families, functional domains, and conserved amino acid residues.

Functional domains identified are :-

InterPro ID	Domain	Significance
IPR001839	Transforming growth factor-beta, C-terminal	Mature biologically active region of MSTN
IPR015616	Growth/Differentiation Factor 8 (GDF8), C-terminal domain	Characteristic domain specific to myostatin
IPR001111	TGF-beta propeptide	Required for protein folding, processing and activation

Conserved motifs identified are :-

InterPro ID	Feature	Function
IPR029034	Cystine-knot cytokine	Stabilizes the protein through conserved cysteine residues
IPR017948	Transforming growth factor beta conserved site	Conserved functional motif involved in TGF- β activity

InterProScan prediction for the *Bos taurus* MSTN protein revealed the presence of six conserved InterPro entries. This protein was annotated as belonging to the Transforming Growth Factor-beta-like (IPR015615) family. It has three functional regions, namely the

TGF-beta propeptide (IPR001111), Transforming Growth Factor-beta C-terminal domain (IPR001839), and

Your InterProScan Search Results (Sequences) ¹

Name

Job ID iprscan5-R20260712-195739-0248-28638501-p2m

InterPro Version 109.0

InterProScan Version 5.78-109.0

Sequence type Amino acids

Number of Sequences 1

Status Finished

Expires Mon Jul 20 2026

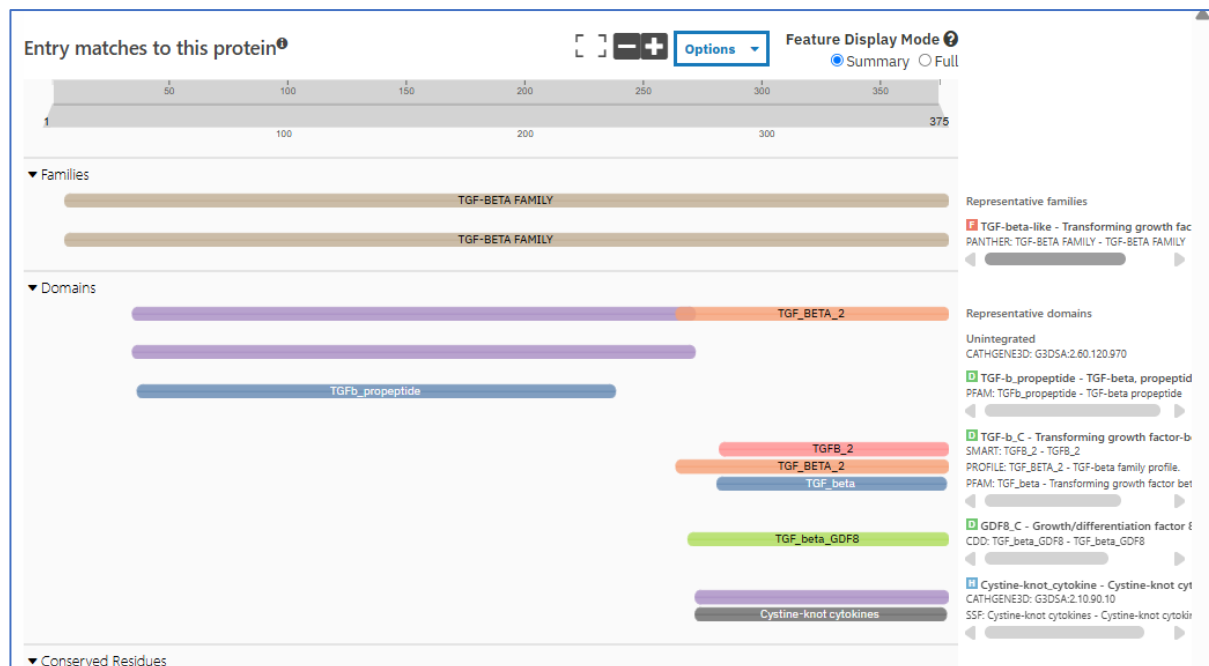
Actions Delete Resubmit Download

1 - 1 of 1 result

Sequence	Matches	Sequence Length
NP_001001525.1	18	375

Previous Next

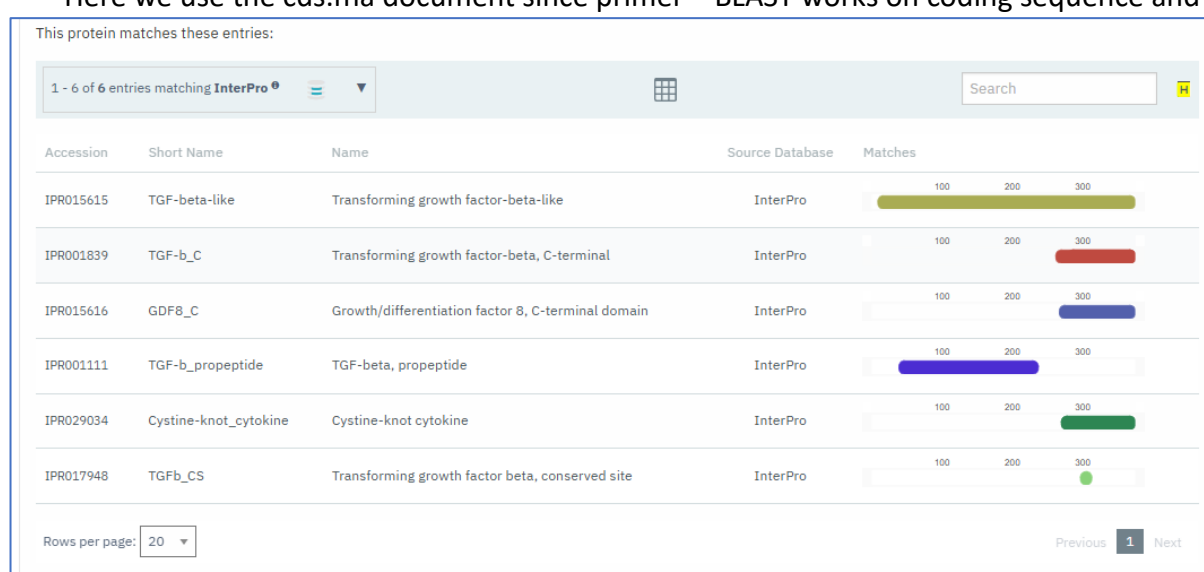
Growth/Differentiation Factor 8 (GDF8) C-terminal domain (IPR015616). In addition, two conserved motifs were identified, that is, the Cystine-knot cytokine motif (IPR029034) and the Transforming Growth Factor-beta conserved site (IPR017948).



These regions are essential for protein folding, structural stability, receptor binding, and signal transduction involved in regulating skeletal muscle growth. The high conservation of these motifs and domains highlights the critical biological role of MSTN and supports its significance in livestock breeding and muscle development studies.

To design gene-specific PCR primers and validate their specificity through in silico PCR analysis.

Here we use the cds.fna document since primer – BLAST works on coding sequence and



the entire gene in gene.fna might contain introns that might not give effective results. PCR requires primers that bind to the target gene and initiate the DNA amplification process. Thus we are designing gene-specific primers to ensure proper amplification of *Bos taurus* MSTN gene. Performing in silico PCR validation before laboratory experiments allows the evaluation of primer specificity, melting temperature, GC content, and expected amplicon size.

Cds.fna was retrieved from NCBI and used as a template for primer design. These primers were designed using NCBI-Primer-BLAST, which combines Primer3 for primer design with BLAST for specificity analysis. Standard primer design parameters were applied, including a primer length of 18–25 nucleotides, a melting temperature (T_m) of 58–62°C, a GC content of 40–60%, and an expected PCR product size of 150–300 bp. Primer specificity was evaluated against the *Bos taurus* RefSeq mRNA database using the integrated Primer-BLAST specificity checking tool.

Input PCR template

NM_001001525.4:134-1261_MSTN_[organism=Bos_taurus]_[GeneID=281187]_[region=cds]

Range

1 - 1128

Your PCR template is highly similar to the following sequence(s) from the search database. To increase the chance of finding specific primers, please review the list below and select all sequences (within the given sequence ranges) that are intended or allowed targets.

Select: All None Selected:1

Accession	Title	Identity	Alignment length	Seq. start	Seq. stop
NM_001001525.4	Bos taurus myostatin (MSTN), mRNA	100%	1128	134	1261

Submit

☐ Show results in a new window

Input PCR template

NM_001001525.4:134-1261_MSTN_[organism=Bos_taurus]_[GeneID=281187]_[region=cds]

Range

1 - 1128

Edit search

Specificity of primers

Primer pairs are specific to input template as no other targets were found in selected database: NCBI Messenger RNA Reference Sequences (Organism limited to Bos taurus)

Other reports

[Search Summary](#)

Graphical view of primer pairs

Query: 1 - 1.1K (1,128 nt)

Primer 1

Primer 2

Primer 3

Primer 4

Primer 5

Primer 6

Primer 7

Primer 8

Primer 9

Primer pair 1

	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	ATGACTACACGCCAGGACG	Plus	20	326	345	61.95	60.00	3.00	3.00
Reverse primer	TGCACAACTGTGCGAGG	Minus	20	515	496	61.71	55.00	5.00	2.00
Product length	190								

Products on intended targets

+NM_001001525.4 Bos taurus myostatin (MSTN), mRNA

product length = 190

Forward primer 1 ATGACTACACGCCAGGACG 20

Template 459 478

Reverse primer 1 TGCACAACTGTGCGAGG 20

Template 648 629

Primer pair 2

	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	ACGATGACTACACGCCAGG	Plus	20	323	342	61.95	60.00	3.00	3.00
Reverse primer	GCACAACTGTGCGAGGA	Minus	20	514	495	61.44	55.00	5.00	1.00
Product length	192								

Products on intended targets

+NM_001001525.4 Bos taurus myostatin (MSTN), mRNA

product length = 192

Forward primer 1 ACGATGACTACACGCCAGG 20

Template 456 475

Reverse primer 1 GCACAACTGTGCGAGGA 20

Template 647 628

Primer pair 3

	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	CTACACGCCAGGACGGAAA	Plus	20	330	349	62.16	60.00	3.00	0.00
Reverse primer	GCTCTGCCAATACCACTGCC	Minus	21	615	595	61.89	57.14	4.00	2.00
Product length	286								

Products on intended targets

+NM_001001525.4 Bos taurus myostatin (MSTN), mRNA

product length = 286

Forward primer 1 CTACACGCCAGGACGGAAA 20

Template 463 482

Reverse primer 1 GCTCTGCCAATACCACTGCC 21

Template 748 728

Primer pair 4

	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	AGAGATGCCAGCTGACGG	Plus	20	292	311	62.34	60.00	4.00	3.00
Reverse primer	GCACAACTGTGCGAGGAG	Minus	21	514	494	62.31	57.14	5.00	0.00
Product length	223								

Products on intended targets

+NM_001001525.4 Bos taurus myostatin (MSTN), mRNA

product length = 223

Forward primer 1 AGAGATGCCAGCTGACGG 20

Template 425 444

Reverse primer 1 GCACAACTGTGCGAGGAG 21

Template 647 627

Primer pair 5									
	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	CGATGACTACCAAGCCAGGA	Plus	20	324	343	61.66	60.00	3.00	1.00
Reverse primer	TACCACTGCTGGGTTCTATGT	Minus	21	604	584	61.39	52.38	6.00	2.00
Product length	281								
Products on intended targets									
*NM_001001525.4 Bos taurus myostatin (MSTN), mRNA									
product length = 281									
Forward primer	1 CGATGACTACCAAGCCAGGA	20							
Reverse primer	1 TACCACTGCTGGGTTCTATGT	21							
Template	457	476							
Reverse primer	1 TACCACTGCTGGGTTCTATGT	21							
Template	737	717							

Primer pair 7									
	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	TTTGCCCAAGGCTCTCCAC	Plus	20	240	259	62.71	60.00	4.00	0.00
Reverse primer	GGGTTTCTCTCCACTTGGGT	Minus	21	405	385	61.36	52.38	2.00	0.00
Product length	166								
Products on intended targets									
*NM_001001525.4 Bos taurus myostatin (MSTN), mRNA									
product length = 166									
Forward primer	1 TTTGCCCAAGGCTCTCCAC	20							
Reverse primer	1 GGGTTTCTCTCCACTTGGGT	21							
Template	373	392							
Reverse primer	1 GGGTTTCTCTCCACTTGGGT	21							
Template	538	518							

Primer pair 8									
	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	CTTTGCCCAAGGCTCTCC	Plus	20	238	257	61.25	60.00	4.00	0.00
Reverse primer	TGGGTTTCTCTCCACTTGGG	Minus	21	406	386	61.36	52.38	3.00	2.00
Product length	169								
Products on intended targets									
*NM_001001525.4 Bos taurus myostatin (MSTN), mRNA									
product length = 169									
Forward primer	1 CTTTGCCCAAGGCTCTCC	20							
Reverse primer	1 TGGGTTTCTCTCCACTTGGG	21							
Template	371	398							
Reverse primer	1 TGGGTTTCTCTCCACTTGGG	21							
Template	539	519							

Primer pair 9									
	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	TCTCGATGCTGTGTTACCTT	Plus	21	835	855	60.95	52.38	4.00	0.00
Reverse primer	ACCTCTGGGGTTTCTTGTT	Minus	20	1002	983	61.66	55.00	5.00	0.00
Product length	168								
Products on intended targets									
*NM_001001525.4 Bos taurus myostatin (MSTN), mRNA									
product length = 168									
Forward primer	1 TCTCGATGCTGTGTTACCTT	21							
Reverse primer	1 ACCTCTGGGGTTTCTTGTT	20							
Template	968	988							
Reverse primer	1 ACCTCTGGGGTTTCTTGTT	20							
Template	1135	1116							

Primer pair 10									
	Sequence (5'→3')	Template strand	Length	Start	Stop	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	GCCAGACGGAACGGTCAT	Plus	20	337	356	62.80	60.00	3.00	3.00
Reverse primer	GGAGTCTTGACAGGCTCAGA	Minus	21	497	477	61.17	57.14	7.00	3.00
Product length	161								
Products on intended targets									
*NM_001001525.4 Bos taurus myostatin (MSTN), mRNA									
product length = 161									
Forward primer	1 GCCAGACGGAACGGTCAT	20							
Reverse primer	1 GGAGTCTTGACAGGCTCAGA	21							
Template	478	489							
Reverse primer	1 GGAGTCTTGACAGGCTCAGA	21							
Template	638	618							

From this report it is concluded that primer 1 has the best overall design.

Parameter of primer 1	Value
Forward primer	ATGACTACCACGCCCCAGACG
Reverse primer	TGCACAAACACTGTGCGAGG
Product size	190 bp

Forward length	20 bp
Reverse length	20 bp
Forward T _m	61.95°C
Reverse T _m	61.71°C
Forward GC	60%
Reverse GC	55%

Ten candidates for primers were generated, primer pair one was selected. The specificity analysis demonstrated that the primer pair uniquely amplified the *Bos taurus* MSTN mRNA (NM_001001525.4), with no additional amplification products detected in the selected *Bos taurus* RefSeq mRNA database, confirming high primer specificity.

Parameter	Forward Primer	Reverse Primer
Sequence (5'→3')	ATGACTACCACGCCCAGACG	TGCACAAACACTGTGCGAGG
Length	20 bp	20 bp
Melting Temperature (T _m)	61.95°C	61.71°C
GC Content	60%	55%
Template Strand	Plus	Minus

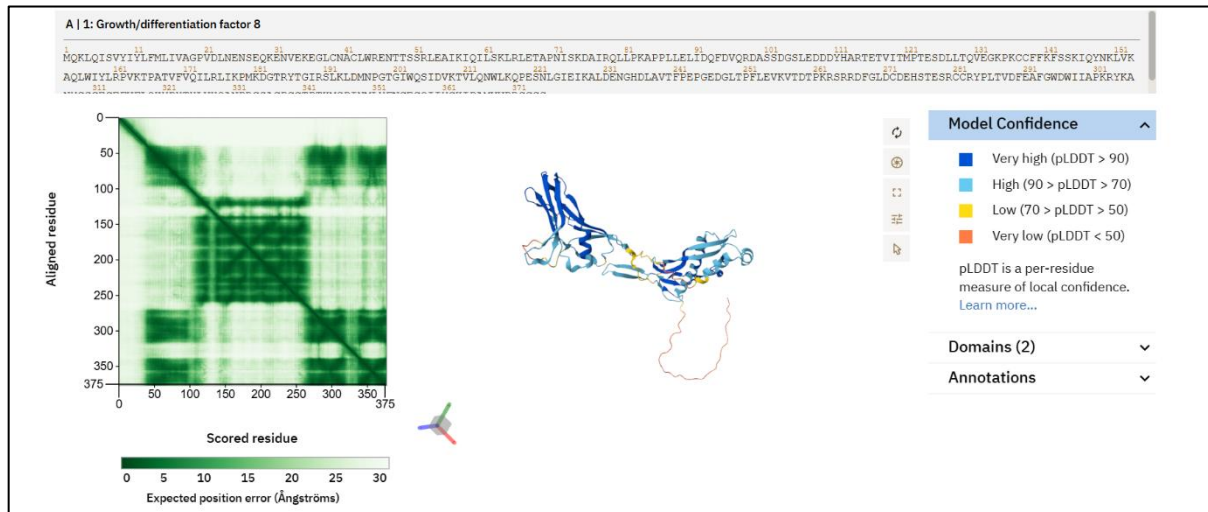
Predicted amplification size is 190 bp.

In silico validation: Primer-BLAST specificity analysis predicted a single PCR product of 190 bp corresponding to the *Bos taurus* MSTN mRNA (NM_001001525.4). No additional amplification products were identified within the selected *Bos taurus* RefSeq mRNA database, indicating that the designed primer pair is highly specific for the target gene and suitable for PCR-based applications.

These primers can be used for applications such as conventional PCR, quantitative PCR (qPCR), gene expression analysis, mutation detection, and molecular characterization of the MSTN gene, which plays a crucial role in regulating skeletal muscle development and has important implications in livestock breeding and animal biotechnology.

To predict, model, and validate the three-dimensional structure of the MSTN protein using state-of-the-art protein structure prediction tools

Proteins are tiny molecular machines. Their 3D shape dictates their job in your body. Visualizing this 3D structure is used to see how they fold, locate their active parts, and understand how they interact with other molecules. Predicting the structure of the *Bos taurus* MSTN protein helps understand the structural basis of its role in regulating skeletal muscle growth.



The amino acid sequence of the *Bos taurus* MSTN (GDF8) protein was retrieved from the UniProt database (Accession: O18836) and analysed using the AlphaFold Protein Structure Database. The predicted model (AF-O18836-F1) was visualized, and model confidence was assessed using the predicted Local Distance Difference Test (pLDDT) confidence score. The Predicted Aligned Error (PAE) plot was also examined to evaluate confidence in the relative positioning of different protein regions.

The generated model consists of 375 amino acid residues and displays the characteristic folded architecture of a member of the Transforming Growth Factor-beta (TGF- β) superfamily.

The pLDDT confidence map showed that the majority of the protein structure is coloured dark blue and light blue, indicating very high and high confidence predictions, respectively. In contrast, a few loop and terminal regions were coloured yellow and orange, representing regions with lower prediction confidence that are likely to be intrinsically flexible or disordered. The Predicted Aligned Error (PAE) plot showed low predicted errors for most structured regions, supporting the reliability of the predicted protein conformation.

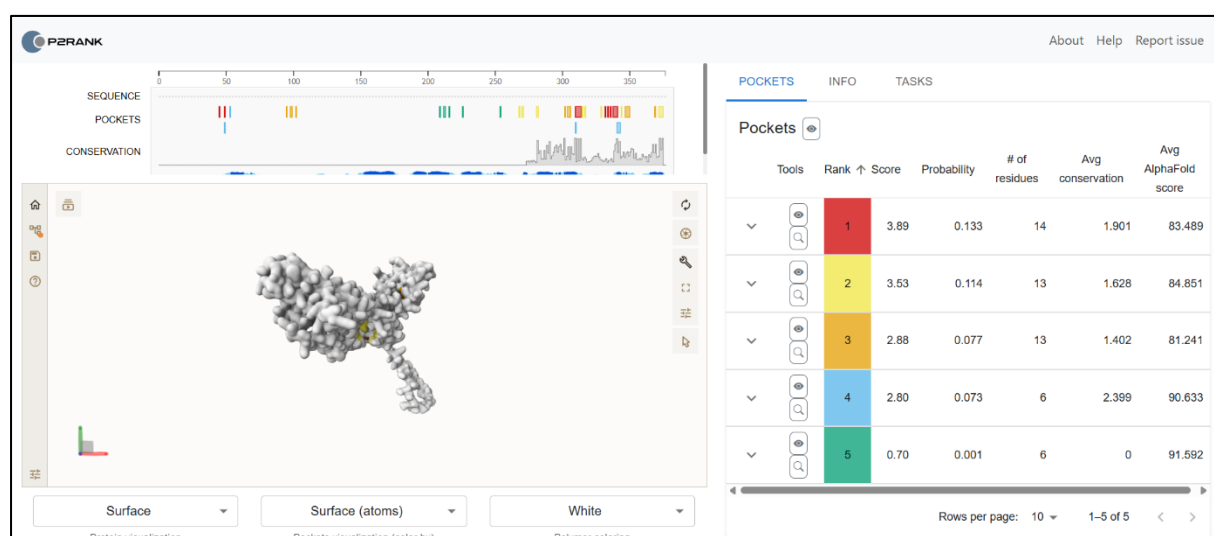
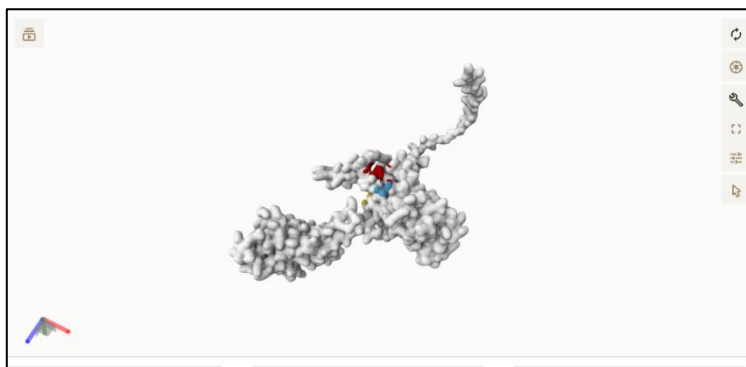
The AlphaFold model indicates that the Bos taurus MSTN protein possesses a well-defined three-dimensional structure with high confidence across most functional regions. The highly confident core structure corresponds to conserved domains responsible for receptor binding and regulation of skeletal muscle growth. The flexible terminal regions with lower confidence are likely involved in protein maturation, activation, or molecular interactions and are commonly observed in signalling proteins.

To identify potential ligand-binding pockets and active sites within the predicted MSTN protein structure.

Finding ligand binding pockets is crucial for drug design, understanding proteins, and disease treatment. These pockets are specific holes or grooves on the surface of a protein (the

"lock") where other molecules (the "keys") fit. Scientists map these pockets to build or find drugs that fit perfectly inside them. This is called structure-based drug design.

The alphafold report is downloaded in pdb format. The accession id 018836 is entered in PrankWeb to visualize the ligand binding sites.



Pocket	Score	Probability	No. of residues	Avg. AlphaFold score
1	3.89	0.133	14	83.49
2	3.53	0.114	13	84.85
3	2.88	0.077	13	81.24
4	2.80	0.073	6	90.63
5	0.70	0.001	6	91.59

Pocket 1 has the highest score of 3.89, binding probability of 0.133, 14 residues which shows that it is the most probable ligand binding site. The average AlphaFold confidence score for this pocket was 83.49, indicating good structural reliability.

Since myostatin acts as a negative regulator of skeletal muscle growth, binding of inhibitory ligands at these sites may alter its biological activity. Identification of these pockets provides suitable target sites for molecular docking studies.

To investigate the interaction of selected bioactive phytochemicals (curcumin, quercetin, and epicatechin) with the MSTN protein through molecular docking analysis.

Molecular docking is a computational method used in drug design to predict how a small molecule (ligand) binds to a larger target protein (receptor). It estimates the binding orientation, intermolecular interactions, and binding affinity between the ligand and protein. In this study, docking was performed to evaluate the potential inhibitory interactions of curcumin, quercetin, and epicatechin with bovine myostatin (MSTN), thereby identifying phytochemicals that may modulate MSTN activity.

The SMILE format of curcumin (CID 969516), Quercetin (CID 5280343) and Epicatechin (CID 72276) was pasted. The results are exported in zip file and attached.

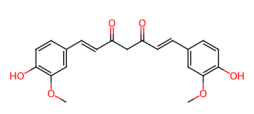
Molecular docking was performed using the SwissDock web server. The three-dimensional structure of MSTN predicted by AlphaFold was used as the receptor, while the three-dimensional structure of curcumin obtained from PubChem served as the ligand. Docking calculations were carried out using default parameters with automatic cavity detection.

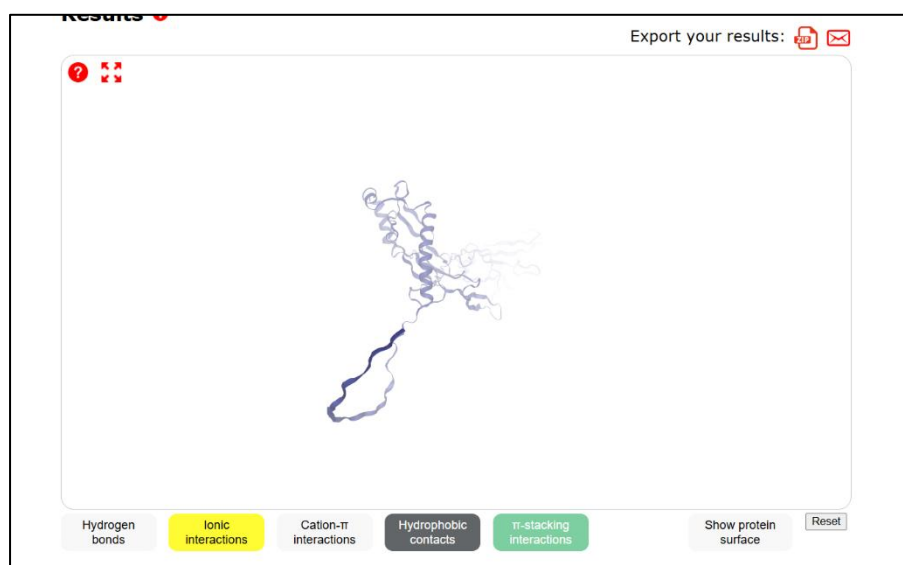
The results of docking with curcumin are as follows: -

Cluster	SwissParam Score
0	-7.5561
1	-7.6122

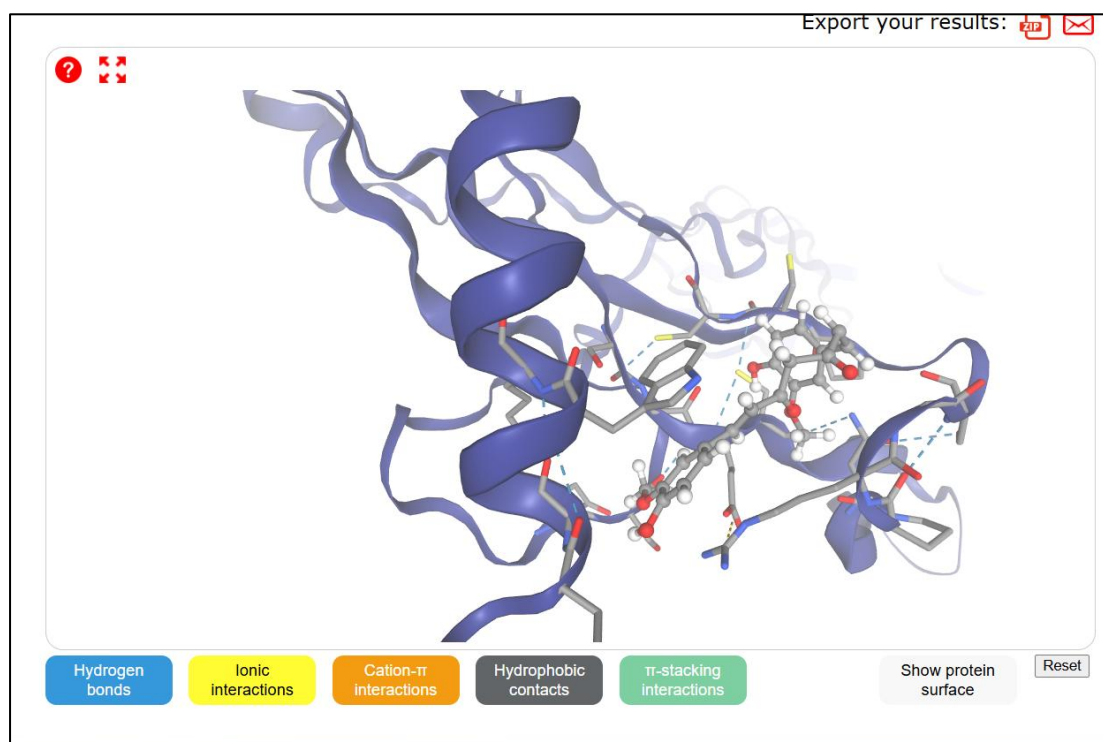
2	-7.3549
3	-7.4523
4	-7.7267
5	-7.3307
6	-7.5090
7	-7.1687
8	-7.8902
9	-7.2270

Cluster 8 has the lowest docking of -7.8902 kcal/mol making it the strongest binding. The best docking pose is the most negative result. This indicates favourable affinity of curcumin to MSTN at cluster 8. This suggests that curcumin can bind stably within a predicted ligand-binding pocket of MSTN through a combination of hydrogen bonding, hydrophobic interactions, and other non-covalent interactions. Since MSTN acts as a negative regulator of skeletal muscle growth, binding of curcumin may influence its activity.

Query		Ligand
Ligand	<chem>COC1=C(C=CC(=C1)/C=C/C(=O)CC(=O)/C=C/C2=CC(=C(C=C2)O)OC)O</chem>	
Target	AF-O18836-F1-model_v6.pdb	
Method	Attracting Cavities 2.0	
Date	July 13, 2026, 5:05 pm UTC	
Parameters: Box center: -2 - -13 - -20 Sampling exhaustivity: medium Number of RIC: 1 Box size: 20 - 20 - 20 Cavity prioritization: buried		
If you publish these results, please, cite the following papers: Bugnon M, Röhrig UF, Goullieux M, Perez MAS, Daina A, Michielin O, Zoete V. SwissDock 2024: major enhancements for small-molecule docking with Attracting Cavities and AutoDock Vina. <i>Nucleic Acids Res.</i> 2024 Röhrig UF, Goullieux M, Bugnon M, Zoete V. Attracting Cavities 2.0: improving the flexibility and robustness for small-molecule docking. <i>J. Chem. Inf. Model.</i> , 2023		



Best members By cluster			
Number of displayed clusters: 10			
Cluster number	Cluster member	AC Score	SwissParam Score
0	1	-0.042102	-7.5561
1	1	0.488506	-7.6122
2	1	0.631658	-7.3549
3	1	0.799165	-7.4523
4	1	1.105097	-7.7267
5	1	1.563543	-7.3307
6	1	2.549001	-7.5090
7	1	4.096384	-7.1687
8	1	4.692527	-7.8902
9	1	5.072456	-7.2270



Here the blue ribbon is the MSTN protein that represents the alpha helices. The stick and ball figure is represented as curcumin ligand. The dashes are the interactions such as hydrogen bonds, ionic interactions, π interactions and hydrophobic contacts. These interactions contribute to the stability of the protein-ligand complex and are consistent with the favourable docking score of -7.89 kcal/mol, suggesting a stable predicted binding conformation.

Molecular docking was performed using the SwissDock web server. The three-dimensional structure of MSTN predicted by AlphaFold was used as the receptor, while the three-dimensional structure of curcumin obtained from PubChem served as the ligand. Docking calculations were carried out using default parameters with automatic cavity detection.

The results of docking with quercetin are as follows: -

Cluster	SwissParam Score
0	-7.4423
1	-7.3864
2	-7.0364
3	-6.7948
4	-6.7320
5	-6.9627
6	-7.1734
7	-6.8692
8	-7.0376
9	-7.1223

Cluster 0 has the lowest docking of -7.4423 kcal/mol making it the strongest binding. The best docking pose is the most negative result. This indicates favourable affinity of quercetin to MSTN at cluster 0. This suggests that quercetin can bind stably within a predicted ligand-binding pocket of MSTN through a combination of hydrogen bonding, hydrophobic interactions, and other non-covalent interactions.

Query

Ligand C1=CC(=C(C=C1C2=C(C(=O)C3=C(C(=C(C=C3O2)O)O)O)O)O

Target AF-O18836-F1-model_v6.pdb

Method Attracting Cavities 2.0

Date July 13, 2026, 5:12 pm UTC

Parameters:

Box center: -2 - -13 - -20 Sampling exhaustivity: medium Number of RIC: 1

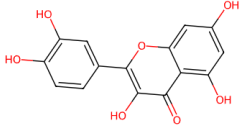
Box size: 20 - 20 - 20 Cavity prioritization: buried

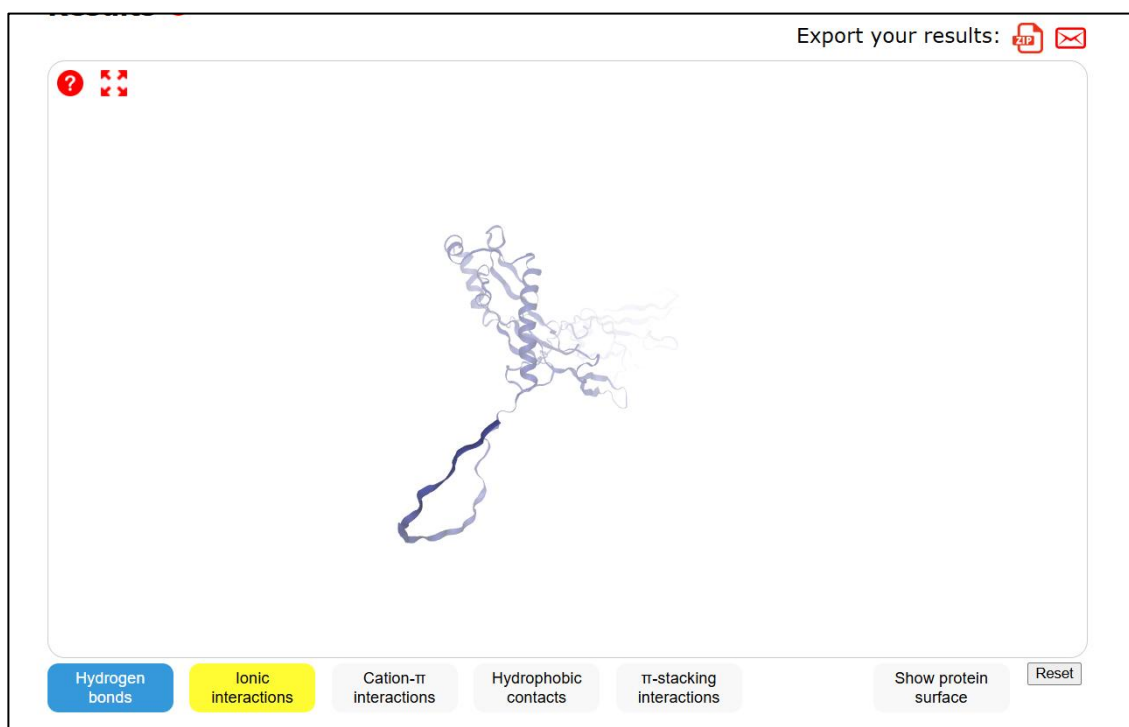
If you publish these results, please, cite the following papers:

Bugnon M, Rohrig UF, Goullieux M, Perez MAS, Daina A, Michielin O, Zoete V. SwissDock 2024: major enhancements for small-molecule docking with Attracting Cavities and AutoDock Vina. *Nucleic Acids Res.* **2024**

Rohrig UF, Goullieux M, Bugnon M, Zoete V. Attracting Cavities 2.0: improving the flexibility and robustness for small-molecule docking. *J. Chem. Inf. Model.*, **2023**

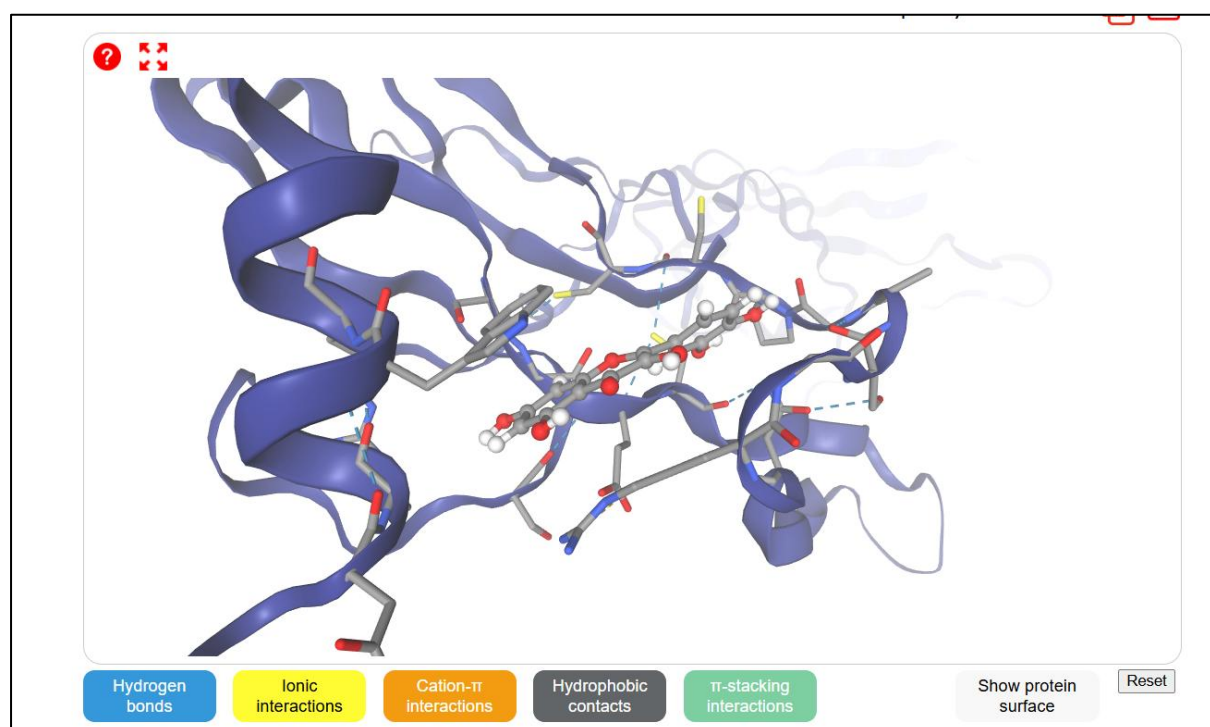
Ligand





Number of displayed clusters: 10

Cluster number	Cluster member	AC Score	SwissParam Score
0	1	2.804165	-7.4423
1	1	4.318576	-7.3864
2	1	5.449803	-7.0364
3	1	6.537359	-6.7948
4	1	7.415069	-6.7320
5	1	7.969773	-6.9627
6	1	9.323963	-7.1734
7	1	9.852782	-6.8692
8	1	10.090129	-7.0376
9	1	10.492277	-7.1223



The molecular docking analysis of quercetin against the predicted *Bos taurus* MSTN protein identified Cluster 0 as the best binding pose, with a SwissParam Score of -7.4423 kcal/mol. The negative docking score indicates a favorable binding affinity between quercetin and MSTN. The docked complex showed that quercetin fits well within the predicted binding pocket of the MSTN protein. Visualization of the protein-ligand complex revealed the presence of several hydrogen bonds, hydrophobic contacts, ionic interactions, cation- π interactions, and π -stacking interactions, which collectively contribute to stabilizing the ligand within the binding cavity.

The results of docking with epicatechin are as follows: -

Cluster	SwissParam Score
0	-6.7908
1	-6.8891
2	-6.7224
3	-6.4698
4	-6.3424
5	-6.7348
6	-6.3199
7	-6.5783
8	-6.8350
9	-6.1702

Cluster 1 has the lowest docking of --6.8891 kcal/mol making it the strongest binding. The best docking pose is the most negative result. This indicates favourable affinity of quercetin to MSTN at cluster 1. This suggests that quercetin can bind stably within a predicted ligand-binding pocket of MSTN through a combination of hydrogen bonding, hydrophobic interactions, and other non-covalent interactions.

Query

Ligand C1[C@H]([C@H](OC2=CC(=CC(=C21)O)O)C3=CC(=C(C=C3)O)O)O

Target AF-O18836-F1-model_v6.pdb

Method Attracting Cavities 2.0

Date July 13, 2026, 5:12 pm UTC

Parameters:

Box center: -2 -13 -20 Sampling exhaustivity: medium Number of RIC: 1

Box size: 20 - 20 - 20 Cavity prioritization: buried

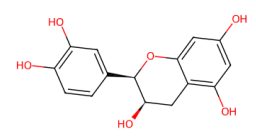
If you publish these results, please, cite the following papers:

Bugnon M, Röhrig UF, Goullieux M, Perez MAS, Daina A, Michielin O, Zoete V. SwissDock 2024: major enhancements for small-molecule docking with Attracting Cavities and AutoDock Vina. *Nucleic Acids Res.* **2024**

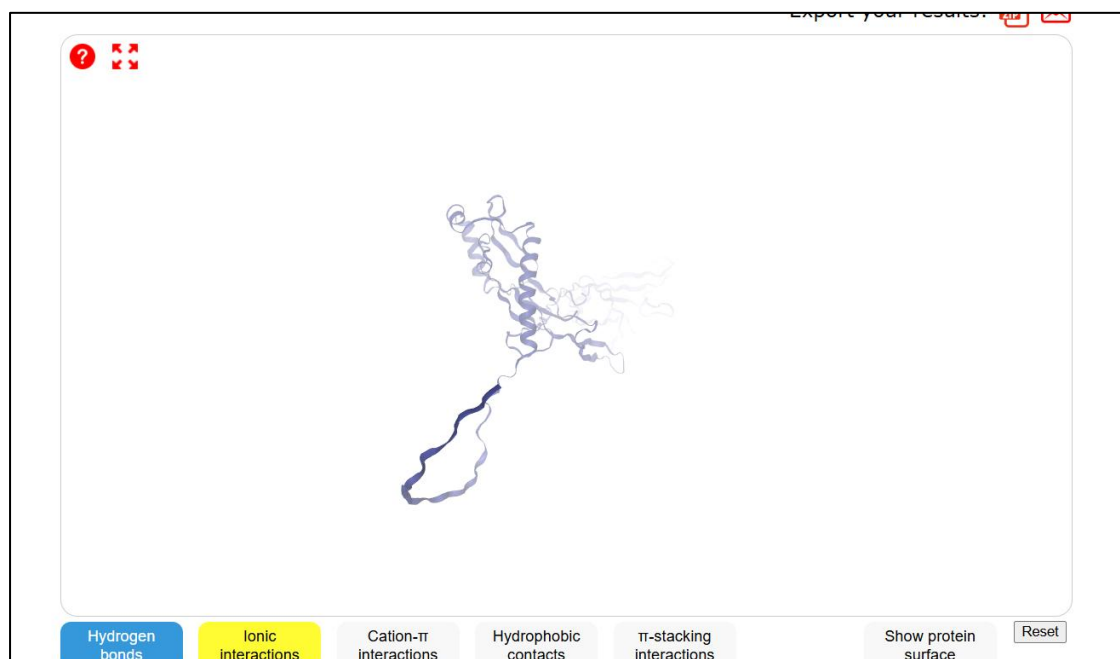
Röhrig UF, Goullieux M, Bugnon M, Zoete V. Attracting Cavities 2.0: improving the flexibility and robustness for small-molecule docking. *J. Chem. Inf. Model.* **2023**

Results

Ligand

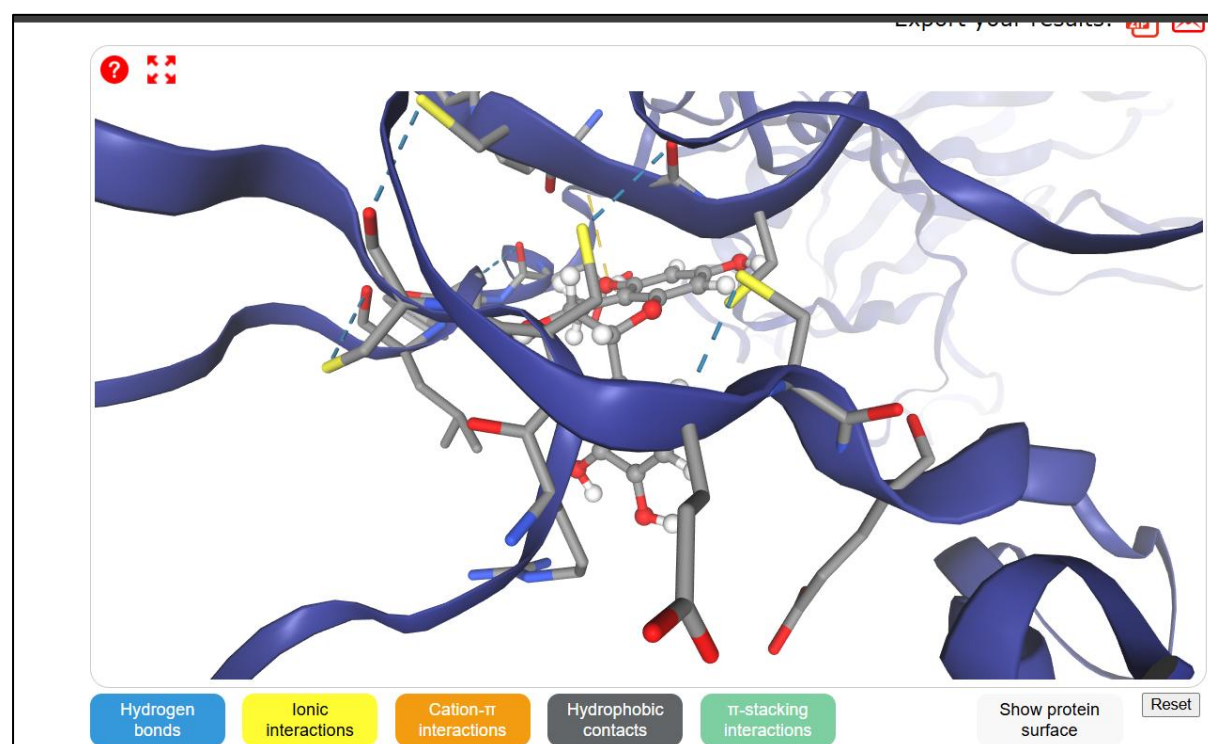


Σ H O P C



Number of displayed clusters: 10 ▾

Cluster number	Cluster member	AC Score	SwissParam Score
0	1	-4.045837	-6.7908
1	1	-3.297671	-6.8891
2	1	-3.190676	-6.7224
3	1	-1.652229	-6.4698
4	1	-0.668267	-6.3424
5	1	-0.649535	-6.7348
6	1	-0.224966	-6.3199
7	1	0.221876	-6.5783
8	1	0.234742	-6.8350
9	1	0.392109	-6.1702



The docking analysis suggests that epicatechin binds within the predicted ligand-binding pocket of MSTN through a combination of hydrogen bonding, ionic interactions, and hydrophobic contacts. These non-covalent interactions help stabilize the ligand within the binding cavity and contribute to the overall binding affinity.

Although the binding affinity of epicatechin is lower than that observed for curcumin and quercetin, the docking results indicate that epicatechin can still interact favourably with MSTN and may influence its biological activity.

All these three ligands are naturally occurring phytochemicals and have well-established antioxidants and anti-inflammatory properties that contribute to the protection and maintenance of skeletal muscle.

Among the three ligands, curcumin exhibited the strongest predicted interaction with MSTN, with the lowest SwissParam Score (-7.8902 kcal/mol). This suggests that curcumin has the highest potential to modulate MSTN activity, which may reduce myostatin-mediated inhibition of muscle growth. Curcumin is also known to suppress inflammatory pathways and oxidative stress, both of which contribute to muscle degeneration. Quercetin demonstrated the second strongest binding affinity (-7.4423 kcal/mol). Epicatechin showed the lowest binding affinity among the three compounds (-6.8891 kcal/mol). Despite its comparatively weaker docking score, its biological properties suggest that it may still contribute to muscle health and could complement other therapeutic approaches targeting MSTN.

To integrate sequence, evolutionary, structural, and docking analyses for assessing the potential functional importance of the MSTN gene in livestock muscle development and its relevance to animal breeding and biotechnology.

The current in-silico study has effectively used the technique of sequence retrieval, comparative genomics, evolution, structure, and molecular docking to conduct an exhaustive analysis of the *Bos taurus* Myostatin MSTN gene and possible functions.

Sequence retrieval and curation was used to validate the sequence identification of the *Bos taurus* MSTN gene and protein sequences available on public biological data bases. CDS and ORF analyses showed that there is an open reading frame in the MSTN coding for the MSTN precursor protein. Basic sequence analysis revealed that the MSTN contains all the typical attributes of TGF- β super family members.

The sequence analysis of MSTN by means of BLAST search, multiple sequence alignment, and phylogenetic analysis revealed that MSTN is a highly conserved gene in livestock animals, such as *Bos taurus*, *Bos indicus*, *Bubalus bubalis*, *Ovis aries*, *Capra hircus*.

The conserved amino acids, conserved motifs, and the functional domains of the protein analysed with InterPro indicate that the structure and the biological function of MSTN have remained unchanged throughout the evolution due to its vital importance in skeletal muscle development regulation.

Also, the study of the genomic organization indicated high conservation of the exon-intron structure of the MSTN gene, which is a reflection of the evolutionary constraint on this gene due to its biological importance.

Gene-specific primer design through Primer-BLAST has enabled successful design of primers that amplify MSTN with an expected product size of 190 base pairs. The in-silico analysis revealed that the designed primers were highly specific, suggesting their applicability to other biological analyses such as gene expression analysis and mutation detection in the future.

The use of AlphaFold Protein Structure Database for structural prediction resulted in the creation of a highly accurate three-dimensional structure of MSTN. It is evident from the high confidence scores of almost all the functional regions of MSTN. Afterward, binding pocket prediction led to the identification of ligand-binding sites, with Pocket 1 being the most probable one.

According to molecular docking analysis, all the three chosen phytochemicals, curcumin, quercetin, and epicatechin had good interactions with the predicted pocket of MSTN. Out of the compounds used in the experiment, curcumin had the highest predicted binding affinity (SwissParam Score: -7.8902 kcal/mol), quercetin had a moderate score (-7.4423 kcal/mol), while epicatechin had the lowest binding affinity (-6.8891 kcal/mol). These interactions were stabilized by various non-covalent interactions like hydrogen bonds, hydrophobic interactions, ionic interactions, cation- π interactions, and π -stacking interactions.

In conclusion, from the computational analyses performed, it is evident that MSTN is a highly conserved and structurally significant molecule that regulates the development of the skeletal muscles in animals. The positive docking interactions between the naturally occurring phytochemicals and MSTN molecule suggest that the phytochemicals can act as modulators for the activity of MSTN. Although docking analysis is just theoretical predictions, the findings obtained will provide good preliminary information for future experiments.

The use of sequence analysis, evolution conservation, structure prediction, and molecular docking shows the practicality of using the bioinformatics approach to study gene functions and identify potential molecular targets useful in the context of animal breeding and livestock improvement.

Conclusion

Results from the current study reveal that the *Bos taurus* MSTN gene is very conserved among livestock species with conservation of key domains critical for muscle regulation. Based on molecular docking, it was observed that curcumin had higher binding affinity towards the MSTN protein than quercetin and epicatechin; therefore, it can be used as a possible phytochemical for further experimental investigations. Thus, this study proves the utility of bioinformatics tools in studying MSTN biology and its possible modulators.

Softwares used

Tool	Purpose
NCBI	Sequence retrieval
ORFfinder	ORF prediction
BLAST	Similarity analysis
Clustal Omega	Multiple sequence alignment
Phylogeny.fr	Phylogenetic tree
InterPro	Domain prediction
Primer-BLAST	Primer design
AlphaFold	Structure prediction
PrankWeb	Pocket prediction
SwissDock	Molecular docking

References

<https://medlineplus.gov/genetics/gene/mstn/>
<https://pubmed.ncbi.nlm.nih.gov/33219121/>
<https://academic.oup.com/edrv/article-abstract/29/5/513/2354982?redirectedFrom=fulltext>
<https://pubmed.ncbi.nlm.nih.gov/30125951/>

Github

<https://github.com/ShaliniRoy2025/MSTN-Bioinformatics-Analysis>



LinkedIn

<https://www.linkedin.com/in/shalini-roy-2859ba331/>

